

BASIC NOTIONS AND RESULTS IN TOPOLOGY

1. METRIC SPACES

A *metric* on a set X is a map $d : X \times X \rightarrow \mathbb{R}_+$ with the properties:

- $d(x, y) \geq 0$ and $d(x, y) = 0 \Rightarrow x = y$,
- $d(x, y) = d(y, x)$,
- $d(x, y) \leq d(x, z) + d(z, y)$,

for all $x, y, z \in X$. The pair (X, d) is called a *metric space*.
Given $Y \subset X$ the *diameter* of Y is

$$\text{diam}(Y) = \sup_{x, y \in Y} d(x, y).$$

Sets with finite diameter are called *bounded* sets. For $x \in X$ and $r > 0$ the set

$$B(x, r) = \{y \in X : d(y, x) < r\}$$

is called the open *ball* centered at x and of radius r and

$$\bar{B}(x, r) = \{y \in X : d(y, x) \leq r\}$$

is called the closed ball centered at x and of radius r .

The set $Y \subset X$ is called *open* if for every $x \in Y$ there exists $r > 0$ such that $B(x, r) \subset Y$. Note that arbitrary unions of open sets are (obviously) open.

The set $Y \subset X$ is called *closed* if $X \setminus Y$ is open. Arbitrary intersections of closed sets are closed. The *closure* of a set $Y \subset X$ is

$$\bar{Y} = \bigcap \{Z \supset Y : Z \text{ is a closed set}\}.$$

The set $Y \subset X$ is called *dense* in X if $\bar{Y} = X$.

If X contains a countable dense subset we say that X is *separable*.

The set $Y \subset X$ is called *nowhere dense* in X if $\bar{Y}$ contains no open ball.

The space X is of the first (Baire) category if it can be written as a countable union of nowhere dense subsets. The space X is of the second (Baire) category if it isn't of the first category.

We say that a sequence (x_n) in X *converges* to $x \in X$ if

$$\lim_{n \rightarrow \infty} d(x_n, x) = 0.$$

If $Y \subset X$ then its closure $\overline{Y}$ consists of Y together with the limits of convergent sequences in Y .

A sequence (x_n) in X is a *Cauchy sequence* if

$$\lim_{m,n \rightarrow \infty} d(x_n, x_m) = 0.$$

Convergent sequences are Cauchy sequences but the converse fails in general.

The metric space (X, d) is called *complete* if every Cauchy sequence is convergent.

Theorem 1.1. *(X, d) is complete if and only if for any decreasing sequence (Y_n) of subsets (i.e. $Y_n \supset Y_{n+1}$) which are nonvoid, closed and satisfy $\lim_{n \rightarrow \infty} \text{diam}(Y_n) = 0$, their intersection consists of exactly one point.*

Proof. Assume the property in the statement holds true and let (x_n) be a Cauchy sequence in X . Set

$$Y_n = \overline{\{x_m : m \geq n\}},$$

and note that these sets are closed, $Y_{n+1} \subset Y_n$, and $\text{diam}(Y_n) \rightarrow 0$. If $\bigcap_{n \geq 1} Y_n = \{x\}$ then

$$d(x_n, x) \leq \text{diam}(Y_n) \rightarrow 0,$$

i.e. (x_n) converges to x . Conversely, assume that (X, d) is complete and let (Y_n) be a sequence as in the statement. Pick $x_n \in Y_n$ and note that if $m \geq n$ then $d(x_n, x_m) \leq \text{diam}(Y_n)$, which implies that (x_n) is Cauchy. Then (x_n) is convergent and its limit must belong to each Y_n since these are closed sets. If x' is another common point of all Y_n then from

$$d(x, x') \leq \text{diam}(Y_n) \quad \forall n$$

we see that $x = x'$. □

Every metric space X can be identified with a dense subset of a complete metric space. The latter can be chosen to be unique up to isometries and is usually called the *completion* of X .

Theorem 1.2. *(Baire) A complete metric space is of the second category.*

Proof. Assume the contrary, that is, X is complete but

$$X = \bigcup_{n=1}^{\infty} Y_n,$$

where Y_n is nowhere dense for all n . Take an arbitrary closed ball $\overline{B}(x_0, r_0)$ with $x_0 \in X$, $r_0 > 0$. Since this ball cannot be contained

in $\overline{Y_1}$, there exists $\overline{B}(x_1, r_1) \subset \overline{B}(x_0, r_0)$ with $r_1 < r_0/2$ such that $\overline{B}(x_1, r_1) \cap Y_1 = \emptyset$. Repeat this procedure inductively, that is, find $\overline{B}(x_2, r_2) \subset \overline{B}(x_1, r_1)$ with $r_2 < r_1/2$ such that $\overline{B}(x_2, r_2) \cap (Y_1 \cup Y_2) = \emptyset$, and so on. We obtain a decreasing sequence of closed balls $\overline{B}(x_n, r_n)$ such that $\overline{B}(x_n, r_n) \cap (\cup_{j=1}^n Y_j) = \emptyset$ and $r_n < r_0/2^n$. By Theorem 1.1 these balls must have a common point which, by construction cannot belong to any set Y_n . Contradiction! $\square$

A subset Y of the metric space (X, d) is called *precompact* or *relatively compact* if every sequence in Y has a convergent subsequence. A closed precompact set is called *compact*. Compact sets are bounded and closed as in $\mathbb{R}^n$, but the converse is far from being true in this generality!

Exercise 1. If X is compact then it is complete and separable.

Theorem 1.3. *A set $Y \subset X$ is compact if and only if from every open cover $\{U_\alpha\}_{\alpha \in A}$ of Y we can extract a finite cover.*

Proof. The same as in the calculus course. $\square$

Let (X, d) , (Z, ρ) be metric spaces. A function $f : X \rightarrow Z$ is called *continuous at $x_0 \in X$* if $\lim_{n \rightarrow \infty} f(x_n) = f(x_0)$ whenever (x_n) is a sequence in X which converges to x_0 . The function f is *continuous on X* if it is continuous at each point of X .

Note that a function $f : X \rightarrow Z$ is continuous on X if and only if the preimage of any open subset of Z is open in X .

Theorem 1.4. *Let (X, d) be a compact metric space. Then every continuous function $f : X \rightarrow \mathbb{R}$ is bounded and attains its minimum and maximum in X .*

Proof. The same as in the calculus course. $\square$

Theorem 1.5. (*uniform boundedness*) *Let (X, d) , (Z, ρ) be metric spaces and let $\mathcal{F}$ be a family of continuous functions from X into Z such that for every $x \in X$ the set*

$$\{f(x) \in Z : f \in \mathcal{F}\}$$

is bounded in Z . If X is complete then there exists a nonvoid open set $U \subset X$ such that

$$\cup_{f \in \mathcal{F}} f(U)$$

is bounded.

Proof. Fix $z \in Z$ and let

$$X_n = \{x \in X : \rho(f(x), z) \leq n \ \forall f \in \mathcal{F}\}.$$

Clearly, each X_n is closed and by assumption we have

$$\bigcup_{n=1}^{\infty} X_n = X.$$

Then by Baire's Theorem (Theorem 1.2) at least one of these sets will contain an open ball. $\square$

Let (X, d) be a compact metric space and let $C(X)$ denote the set of all complex-valued continuous functions on X . Given $f, g \in C(X)$ set

$$\rho(f, g) = \sup_{x \in X} |f(x) - g(x)|.$$

Exercise 2. Show that $(C(X), \rho)$ is a complete metric space and that a sequence converges in this space if and only if it converges uniformly on X .

A bounded set in $(C(X), \rho)$ is sometimes called a *uniformly bounded family* of functions.

A subset $\mathcal{F} \subset C(X)$ is called *equicontinuous* at $x \in X$ if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$|f(x) - f(y)| < \varepsilon$$

whenever $d(x, y) < \delta$ and $f \in \mathcal{F}$! The set $\mathcal{F} \subset C(X)$ is called equicontinuous if it is equicontinuous at all $x \in X$.

Theorem 1.6. (*Arzela-Ascoli*) Let (X, d) be a compact metric space. If $\mathcal{F} \subset C(X)$ is bounded and equicontinuous then every sequence in $\mathcal{F}$ has a convergent subsequence in $C(X)$ (i.e. $\mathcal{F}$ is precompact in $C(X)$).

Proof. Let (f_n) be a sequence in $\mathcal{F}$. Recall that X is separable and fix a countable dense subset $\{x_1, \dots, x_n, \dots\}$. Since $(f_n(x_1))$ is a bounded sequence of complex numbers, it has a convergent subsequence $(f_{n_k^1}(x_1))$. At its turn $(f_{n_k^1}(x_2))$ has a convergent subsequence indexed by (n_k^2) and by construction, $(f_{n_k^2}(x_j))$ converges in $\mathbb{C}$ for $j = 1, 2$. Inductively we construct ("smaller and smaller") subsequences (n_k^m) such that $(f_{n_k^m}(x_j))$ converges in $\mathbb{C}$ for $1 \leq j \leq m$. Then the "diagonal" subsequence $(f_{n_k^k})$ converges pointwise on the set $\{x_1, \dots, x_n, \dots\}$.

Let $\varepsilon > 0$. Since $\mathcal{F}$ is equicontinuous, for every $x \in X$ there exists $\delta_x > 0$ such that

$$d(x, y) < \delta_x \Rightarrow |f(x) - f(y)| < \varepsilon, \quad \forall f \in \mathcal{F}.$$

Clearly,

$$X = \bigcup_{x \in X} B(x, \delta_x),$$

and since X is compact there exist $z_1, \dots, z_n \in X$ such that

$$X = \bigcup_{j=1}^N B(z_j, \delta_{z_j}).$$

For $1 \leq j \leq N$ choose $x_{n_j} \in B(z_j, \delta_{z_j})$ and note that if $x \in B(z_j, \delta_{z_j})$ then

$$\begin{aligned} |f_{n_k^k}(x) - f_{n_l^l}(x)| &\leq |f_{n_k^k}(x) - f_{n_k^k}(x_{n_j})| + |f_{n_k^k}(x_{n_j}) - f_{n_l^l}(x_{n_j})| \\ &\quad + |f_{n_l^l}(x_{n_j}) - f_{n_l^l}(x)| \end{aligned}$$

Now the first and the third term on the right hand side are always less than ε . By the previous argument we conclude that there exists k_ε such that if $k, l \geq k_\varepsilon$ and $1 \leq j \leq N$,

$$|f_{n_k^k}(x_{n_j}) - f_{n_l^l}(x_{n_j})| < \varepsilon,$$

which leads to

$$|f_{n_k^k}(x) - f_{n_l^l}(x)| \leq 3\varepsilon$$

for all $x \in X$ and the result follows. $\square$

2. TOPOLOGICAL SPACES

A *topology* on a set X is a collection τ of subsets of X with the properties:

- $\emptyset, X \in \tau$,
- Finite intersections of elements of τ belong to τ ,
- Arbitrary unions of elements of τ belong to τ .

(X, τ) is called a topological space and the subsets of X which belong to τ are called *open sets*.

A *neighborhood* of a point $x \in X$ is an open set containing x .

A *basis of neighborhoods* is a collection $\tau' \subset \tau$ such that every open set is a union of elements of τ' .

The topology τ on X is called a *Hausdorff* topology if for every $x, y \in X$, $x \neq y$ there exist open sets $U \ni x$, $V \ni y$ with $U \cap V = \emptyset$.

A point $x \in X$ is a *point of contact* of the set $Y \subset X$ if every neighborhood of x intersects Y .

A set $Y \subset X$ is *closed* if $X \setminus Y$ is open. The closure $\bar{Y}$ of Y is the smallest closed set containing Y and it equals the union of Y and all its points of contact.

If (X, τ) , (Z, η) are topological spaces, a function $f : X \rightarrow Z$ is continuous if $f^{-1}(U)$ is open in X whenever U is open in Z .

Example Let X be a set, let $\{(X_\alpha, \tau_\alpha)\}_{\alpha \in A}$ be a family of topological spaces and let $\{f_\alpha\}_{\alpha \in A}$ be a family of functions with $f_\alpha : X \rightarrow X_\alpha$. Then the smallest topology τ on X for which each function f_α , $\alpha \in A$ is continuous is given by the basis of neighborhoods

$$V = \bigcap_{i=1}^n f_{\alpha_i}^{-1}(U_i),$$

where $n \in \mathbb{N}$, $\alpha_1, \dots, \alpha_n \in A$ and $U_i \in \tau_{\alpha_i}$ are arbitrary.

A concrete example of this type is the product topology. Let $\{(X_\alpha, \tau_\alpha)\}_{\alpha \in A}$ be a family of topological spaces and consider their cartesian product

$$X = \prod_{\alpha \in A} X_\alpha,$$

which is the set of all functions

$$f : A \rightarrow \bigcup_{\alpha \in A} X_\alpha \quad \text{with } f(\alpha) \in X_\alpha \quad \forall \alpha \in A.$$

Such functions are also called *tuples* and denoted by $(x_\alpha)_{\alpha \in A}$, where we understand that $x_\alpha \in X_\alpha$ for all $\alpha \in A$. The product topology is the smallest topology (on the product space) with the property that all "projections"

$$P_\beta : \prod_{\alpha \in A} X_\alpha \rightarrow X_\beta, \quad P_\beta((x_\alpha)_{\alpha \in A}) = x_\beta,$$

are continuous. According to the above observation, a basis of neighborhoods for the product topology is given by the sets

$$V = \{(x_\alpha)_\alpha \in A : x_{\alpha_i} \in U_i\},$$

where $n \in \mathbb{N}$, $\alpha_1, \dots, \alpha_n \in A$ and $U_i \in \tau_{\alpha_i}$ are arbitrary.

The topological space (X, τ) is compact if every open cover of X contains a finite cover. Equivalently, $(X; \tau)$ is compact if any family of closed subsets of X with the property that any finite subfamily has a nonvoid intersection, must have itself a nonvoid intersection.

A family of arbitrary subsets of X with the above property, that is, any finite subfamily has a nonvoid intersection is said to have the *finite intersection property*.

Exercise 3. Show that (X, τ) is compact if and only if every family of subsets of X with the finite intersection property, has at least a common point of contact.

Theorem 2.1. (*Tychonov*) *The cartesian product of compact topological spaces is compact with respect to the product topology.*

Proof. We prove that the condition in Exercise 3 holds. Let $\mathcal{M}$ be a family of subsets of $\prod_{\alpha \in A} X_\alpha$ with the finite intersection property. Use Zorn's lemma to conclude that there exists a maximal family $\mathcal{M}_0$ which has the finite intersection property and contains $\mathcal{M}$. By maximality it follows that $\mathcal{M}_0$ has the following properties:

- (a) Any finite intersection of sets in $\mathcal{M}_0$ belongs to $\mathcal{M}_0$,
- (b) A set that intersects any finite collection of subsets of $\mathcal{M}_0$ must belong to $\mathcal{M}_0$.

To see (a) note that if A is a finite intersection of sets in $\mathcal{M}_0$ then $\mathcal{M}_0 \cup \{A\}$ has the finite intersection property and contains $\mathcal{M}$, hence, it must equal $\mathcal{M}_0$. The proof of (b) is identical; if B satisfies that property $\mathcal{M}_0 \cup \{B\}$ contains $\mathcal{M}$ and has the finite intersection property, hence, it must equal $\mathcal{M}_0$.

Given a set $S \in \prod_{\alpha \in A} X_\alpha$ and $\alpha \in A$, let

$$S_\alpha = \{x_\alpha : (x_\beta)_{\beta \in A}\}$$

be the projection of S on the α coordinate. Let $\mathcal{M}_0^\alpha$ the family of such projections of all sets in $\mathcal{M}_0$. Clearly, $\mathcal{M}_0^\alpha$ has the finite intersection property and since X_α is compact, it follows that $\mathcal{M}_0^\alpha$ must have a common point of contact $\bar{x}_\alpha \in X_\alpha$. Set $\bar{x} = (\bar{x}_\alpha)_{\alpha \in A}$ and consider a neighborhood V of $\bar{x}$ as given in our discussion of the product topology, that is, a finite intersection of sets of the form

$$V_{\alpha_i} = \{(x_\alpha)_{\alpha \in A} : x_{\alpha_i} \in U_i\}, \quad i = 1, \dots, n,$$

where U_i is an open subset of X_{α_i} for all i . Since each U_i intersects every set in $\mathcal{M}_0^{\alpha_i}$ it follows that each V_{α_i} intersects every set in $\mathcal{M}_0$. Then by (b) $V_{\alpha_i} \in \mathcal{M}_0$ and by (a) we obtain that $V \in \mathcal{M}_0$. Thus by the finite intersection property V intersects every set in $\mathcal{M}_0$, in particular every set in $\mathcal{M}$. Since the sets V as above form a basis of neighborhoods we conclude that $\bar{x}$ is a point of contact for every set in $\mathcal{M}$ and we are done. $\square$